

Evgeniia Tokarchuk, PhD

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Summary

NLP and machine learning researcher specializing in representation geometry, neural machine translation, retrieval-augmented models, and efficient modeling.

Work Experience

- 05/2021 – 03/2026  **PhD Researcher.** University of Amsterdam. 
- ▶ Led end-to-end **NLP research** (problem → publication), delivering 4+ first-author papers (EMNLP, TMLR, NAACL, EACL)
 - ▶ Designed and ran **large-scale experiments** using **PyTorch** on **Slurm HPC**
 - ▶ Introduced geometry-aware **optimization** improving **retrieval** quality and decoding **efficiency**, and presented results at international venues.
- 10/2019 – 03/2021  **Research Assistant (Machine Learning & NLP).** eBay Aachen 
- ▶ Designed and implemented **end-to-end training** for pivot-based neural machine translation using non-autoregressive **Transformer** models.
 - ▶ Applied **reinforcement learning** techniques to optimize sequence-level objectives in pivot-based translation systems.
- 02/2018 – 09/2019  **Research Assistant (Machine Learning & NLP).** Machine Learning and Human Language Technology Lab, RWTH Aachen University 
- ▶ Built and optimized NER data and training pipelines in Flair (**PyTorch**) and conducted large-scale character- and subword-level NMT experiments with RETURNN (TensorFlow) on **HPC infrastructure**.
- 10/2016 – 08/2017  **Algorithms Developer.** PressFoto R&D department 
- ▶ Worked on embedding-based similarity search to improve image **retrieval** quality for text-driven queries.
 - ▶ Optimized ranking and matching algorithms, focusing on relevance and retrieval precision in large-scale media datasets.
- 05/2016 – 08/2016  **Google Summer of Code Intern.** The HoneyNet project 
- ▶ Designed and implemented SNARE and TANNER honeypot systems from scratch using **Python** and aiohttp.
 - ▶ Built scalable, asynchronous services for collecting and analyzing malicious network activity.
 - ▶ Delivered production-quality open-source contributions adopted by the security research community.

Skills

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|-----------------------------|--|
| Machine Learning & Research |  Representation learning, transformer models, embeddings geometry, optimization, evaluation, data pipeline, retrieval methods |
| Technical |  Python (10+yrs), PyTorch, fairseq, Linux, bash shell, git, slurm, faiss |
| Languages |  Strong reading, writing and speaking competencies for English (C1), Russian (native), Dutch (A2) |
| Misc. |  Academic research, teaching, $\LaTeX$ typesetting and publishing. |

Education

- 05/2021 – 05/2026  **Ph.D., University of Amsterdam** Machine Learning & NLP.
Thesis title: *Representation collapse and where to find it in Transformer-based Neural Machine Translation* 
- 10/2017 – 03/2021  **M.Sc. Software Systems Engineering, RWTH Aachen University**
Thesis title: *Tight model integration for pivot-based Neural Machine Translation*.
- 09/2011 – 06/2016  **Dipl. Information Security for Automated Systems, South Ural State University**
Thesis title: *Software Package for the Security of Mobile devices*.

Selected Research Publications

- 1** **Evgeniia Tokarchuk** and Vlad Niculae. “The Unreasonable Effectiveness of Random Target Embeddings for Continuous-Output Neural Machine Translation”. In: *Proceedings of the Association for Computational Linguistics: NAACL 2024*.  DOI: 10.18653/v1/2024.naacl-short.56.
Keywords: embedding geometry, representation learning, learning in continuous space.
- 2** **Evgeniia Tokarchuk**, Sergey Troshin, and Vlad Niculae. “Angular Dispersion Accelerates k -Nearest Neighbors Machine Translation”. In: *Findings of the Association for Computational Linguistics: EMNLP 2025*.  DOI: 10.18653/v1/2025.findings-emnlp.759.
Keywords: dispersion, machine translation, approximate k -nearest neighbors search, vector retrieval, representation geometry, retrieval efficiency.
- 3** **Evgeniia Tokarchuk** et al. “Representation Collapse in Machine Translation Through the Lens of Angular Dispersion”. In: *Findings of the Association for Computational Linguistics: EACL 2026*.  DOI: 10.18653/v1/2026.findings-eacl.126.
Keywords: embedding geometry, representation collapse, neural machine translation, transformer models, dispersion.
- 4** **Evgeniia Tokarchuk**, Hua Chang Bakker, and Vlad Niculae. “Keep your distance: learning dispersed embeddings on S_m ”. In: *Transactions on Machine Learning Research* (2025).  URL: <https://openreview.net/forum?id=5JIQE6HcTd>.
Keywords: optimization, representation learning, hyperspherical embeddings, dispersion.

Miscellaneous Experience

Awards and Scholarships

- 2018  **DAAD Study Scholarships for Graduates of All Disciplines**

Teaching and Community Service

- 04/2022 – 06/2024  **Teaching Assistant, NLP 2.** University of Amsterdam.
-  **M.Sc and B.Sc. thesis supervisor.** University of Amsterdam.
- 2021 – now  **Conference Reviewer.** ACL*, NeurIPS, ICML, ICLR
- 03/2023 – now  **Chairwoman.** Stichting “Volunteers with no limits” 
Lead strategic and operational coordination of a volunteer-run charity supporting Ukraine, overseeing donation campaigns, partnerships, and distribution efforts